Bike Rental Writeup

This bike rental project is my first true solo data engineering project. So, I kept the central goal of the entire project in mind "A bike rental company has asked you to create a database to help their analysts understand the effects of weather on bike rentals." (Codecademy). The data provided for this project is a 2016 weather data CSV from Newark, New Jersey, U.S.A, and bike rental data in the form of 12 CSVs (one CSV per month) from Citi Bike.

Firstly, in python I cleaned and transformed the data appropriately. I dropped two columns in the weather dataframe (PGTM & TSUN) because they were null in all rows. I also dropped nulls in the bike rental dataframe where column is "User Type", because the missing data was very small, only 380 rows, while there were still 247,204 rows remaining. Connecting to the SQL database with pgAdmin4 was fairly successful, but because I have never done it in python ChatGPT was used to guide me (I commented where in my Jupyter Notebook). I had to rename all columns in lower-case, so that no errors occurred when sending data to the SQL database.

The second order of business was to create appropriate database table relationships. Firstly, I kept the weather data in a complete table. Secondly, I split the bike rental data into two tables. The 1st table was called "*main_bike_data*", and had columns: main_id (primary key), trip_duration, bike_id, user_type, birth_year, and gender. The 2nd table was named "*bike_trip_data*" and had columns: trip_id (primary_key), main_id (foreign_key), start_time, stop_time, start_station_id, start_station_name, start_station_latitude, start_station_longitude, end_station_id, end_station_name, end_station_latitude, and end_station_longitude. Though admittedly, I ran into problems with this table schema. I had a fundamental mis-conception that

PostgreSQL would create foreign key numbers for me whilst not understanding I had to generate them in python myself or have an already dedicated foreign key column present in the original dataset. So, after understanding more, I chose to keep all of the bike rental data together into one relational table. The new schema had a one-to-many relationship from the weather data table to the bike data table.

In conclusion to the project, I created three SQL data views after having both tables set up properly in PgAdmin4. The data views answered questions: How many bike riders are there a day accounting for the collective trip durations as well? What amount of bike riders are ‘Subscribers’ rather than ‘Customers’? And lastly, how do weather conditions (temperature average, precipitation, and wind speed averages) affect total trips a day? These questions are simply general prompts that may be useful for the analyst teams.